Himanshu Nakrani

+91-823-845-5702 | himanshunakrani0@gmail.com | Hyderabad, India | LinkedIn | GitHub | Portfolio

Professional Summary

AI Engineer and technical lead with over 3 years of experience architecting and deploying production-grade Generative AI and NLP applications, agentic AI systems, and RAG pipelines. Proven expertise in fine-tuning large language models, developing sophisticated Text-to-SQL systems, and engineering scalable AI inference systems that drive significant performance and efficiency improvements for enterprise AI solutions.

Technical Skills

Generative AI & LLMs: Large Language Models, RAG (Retrieval-Augmented Generation), Agentic AI, Multi-agent Systems, AI Guardrails, Responsible AI, Text-to-SQL, Fine-tuning (LoRA/PEFT), TRL, Prompt Engineering, Embeddings, Model Optimization, LangChain, LangGraph, Hugging Face, OpenAI API, Azure OpenAI, Google ADK
Backend Development: FastAPI, REST APIs, SQLAlchemy, Microservices, API Design
Databases: SQL, Vector Databases (pgvector), Semantic Search, Query Optimization
Machine Learning: Deep Learning, NLP, Transformers, PyTorch, JAX/MaxText, Data Pipelines, Model Deployment, Data Augmentation
Cloud Platforms: Azure, AWS, Google Cloud TPU, Oracle Cloud Infrastructure (OCI), Databricks
Programming Languages: Python, Java, SQL, Shell Scripting
Developer Tools: Git, Docker, Harness, CI/CD, MLOps, LLMOps, Model Observability, Testing, Evaluation Frameworks

Professional Experience

State Street Corporation — Hyderabad, India
Emerging Lead - AI Engineer — *July 2023 – Present*

- Architected and deployed "**Alpha Copilot**", an enterprise LLM-powered Text-to-SQL chatbot enabling natural language queries over structured financial data, serving 100+ internal users
- Engineered scalable AI inference infrastructure using **FastAPI and SQLAlchemy**, implementing advanced caching and **latency optimization** techniques that reduced average response time by **75% (from 25-30s to 6-8s)**
- Developed production-ready **RAG system** for extracting actionable insights from fund prospectuses, engineering an optimized document pipeline with intelligent chunking, semantic embeddings, and vector database integration achieving sub-second retrieval times
- Integrated LLM-based data visualization capabilities, automatically generating interactive charts and summaries from query results, improving data accessibility and decision-making speed
- Led backend development for **WealthAI application**, building real-time portfolio narrative generation system that processes optimizer outputs and delivers personalized investment insights at scale
- Fine-tuned **GPT-4o mini, GPT-4.1 mini** (Azure OpenAI) and **Llama 3.1-8B** (Databricks) using supervised fine-tuning and PEFT/LoRA, managing end-to-end LLMOps pipeline including data augmentation, training, and inference optimization
- Conducted research on advanced Text-to-SQL generation techniques, exploring schema-linking methods and few-shot prompting strategies to improve SQL accuracy by 25%
- Spearheaded development of "**Agent Forge**", an enterprise **agentic AI** platform centralizing agent registration, discovery, and governance — actively used by **5+ business teams** with **50+ registered agent workflows**, providing a unified inventory of metadata, capabilities, and performance indicators across **multi-agent systems**
- Designed and implemented OTEL-compliant **model observability**, advanced **AI guardrails** with input/output validation and content filtering, and automated evaluation pipelines for **Responsible AI** compliance across all registered agents; building a no-code agentic workflow builder for visually designing and deploying custom AI agents
- Achieved **95%+ unit test coverage** and streamlined deployment workflows by architecting automated **CI/CD pipelines** with **Harness** and **Docker**, significantly improving system reliability and reducing production incidents
- Collaborated with cross-functional teams in an Agile environment, partnering with business analysts and compliance officers to align deliverables with regulatory requirements and business objectives

State Street Corporation — Hyderabad, India
Software Development Intern — *January 2023 – June 2023*

- Led migration of legacy ideation platform from .NET Core to modern **FastAPI + ReactJS** stack, improving application performance by 40% and developer productivity
- Designed and implemented RESTful APIs with SQLite backend, ensuring seamless frontend integration and comprehensive test coverage

Publications & Research

GoT4SQL-DA: A Graph-of-Thoughts Framework for Scalable Text-to-SQL Data Augmentation FLLM 2025
GRAFT: Graph-of-Thoughts based Reasoning and Augmentation for Finetuning Text-to-SQL CSCI 2025

Education

Nirma University Ahmedabad, India
Bachelor of Technology in Electronics and Communication Engineering *June 2019 – July 2023*

Key Projects

TinyMathReason-1B: Mathematical Reasoning LLM | *JAX/MaxText, PyTorch, TRL, Cloud TPU, AMD MI300X, GCS, Python* | GitHub

- Pretrained a **1.12B parameter** Llama-style decoder-only transformer from scratch on a Google Cloud **TPU v4-32** cluster using **MaxText/JAX**, processing a **300B token** math-heavy dataset with a throughput of **66 TFLOP/s/device**
- Engineered a distributed data pipeline across parallelized cloud instances, processing **4 large-scale datasets** (FineWeb-Edu, OpenWebMath, Proof-Pile-2, Stack-Edu) with MinHash deduplication and sequence packing into optimized binary shards on GCS
- Built custom JAX/Orbax to Hugging Face checkpoint conversion utilities and implemented a two-stage post-training curriculum (**SFT, DPO/GRPO**) using **TRL** on an **AMD MI300X** GPU with custom tokenizer extensions for chain-of-thought reasoning

CogniDoc: RAG Document Q&A Platform | *FastAPI, Supabase pgvector, OpenAI, Gemini, Vercel* | Live

- Developed a production-ready **Retrieval-Augmented Generation (RAG)** pipeline encompassing document ingestion, intelligent chunking, and context-aware semantic retrieval utilizing **Supabase pgvector**
- Engineered a scalable **FastAPI** backend with multi-model LLM routing (**OpenAI, Gemini**) to optimize response latency and relevance, empowering real-time conversational memory in a **Vercel**-deployed application

Aegis: Agent Evaluation & Guardrails Harness | *Google Gemini + ADK, FastAPI, PostgreSQL* | GitHub

- Built a production harness for agentic systems featuring evaluation frameworks, guardrails engine, and observability to ensure reliability of multi-agent workflows
- Leveraged Google ADK primitives (SequentialAgent, ParallelAgent, LoopAgent) to implement structured multi-agent patterns with shared state management
- Developed LLM-as-Judge evaluation and policy-based guardrails with real-time tracing, enabling safe and measurable deployment of agentic applications

Certifications

AWS Certified AI Practitioner | View Certificate
Oracle Cloud Infrastructure Certified Generative AI Professional | View Certificate